What a Repair Leaves Behind: Evaluating Meaning-Preservation Interventions for LLM Sentence Expansion in AAC with Benefit, Harm and Cost Together

Anonymous submission

Abstract

Augmentative and alternative communication (AAC) apps increasingly let a large language model expand a few selected words into a sentence spoken on the user's behalf. Fixes for meaning errors in that sentence are usually reported by their targeted gain alone. We evaluate interventions offline with a ledger that reports benefit, harm and cost together, credits each intervention only with errors it can act on, and fixes a kill criterion in advance. A runtime guard that checks the cues the user supplied and regenerates repaired 63/80 scored violations on four LLMs, raising strict pass from 94.25% to 98.78% at 0.104 extra calls per input. It left 17 violations in place, and a pass is only partial: one scored repair is the fragment "Ask you to wear headphones." Filling missing readings into a three-suggestion menu raised sets offering both polarity or both certainty readings from 15/80 to 78/80, while politeness coverage fell by 35.0 points. The same diversification is the wrong remedy once the input is supplied: for typed sentences, three-suggestion menus contained an output whose label departs from a supplied value in 62.76% of sets, versus 2.70% for three samples. On 78 fresh speech-act item families, a user-selected cue marking the sentence type raised preservation by 31.41 points (Holm $p = .0003$), yet reached only 73.24% against 99.20% for the full sentence, failing our own $\geq$80%-of-full criterion. Labels come from an LLM extractor that agrees with two human annotators at held-out $\kappa = 0.91$–0.97 on untreated outputs; repaired text was not human-rated, and we measure no human effects. We propose, but have not run, a micro-randomized protocol to test whether these offline ledgers predict effects for AAC users.

1 Introduction

People who use AAC often compose by selecting a few words or symbols, because every selection costs time and effort (Valencia et al. 2023; Cai et al. 2024). LLM keyword expansion turns `want go party` into a full sentence (Shen et al. 2022; Cai et al. 2024), which is then spoken as the user's own (Hancock, Naaman, and Levy 2020; Valencia et al. 2020). The model chooses more than words: whether something happened or did not, whether the sentence asks or tells, and whether it says "please".

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Interventions can reduce wrong choices. A system can check its output against what the user supplied and try again (Madaan et al. 2023; Gou et al. 2024); it can offer suggestions that span the readings the keywords leave open; or the user can add a cue such as `[NOT]` (Valencia et al. 2024; Mao et al. 2025; Xiang et al. 2025). Such interventions are usually reported by the gain on the endpoint they target. That is not enough when the output speaks for a person. A repair can fix a dropped negation and also rewrite a sentence that was already right; a menu can gain a missing reading by giving up one the user needed. Offered wording matters: suggested text shifts what people write and believe (Arnold, Chauncey, and Gajos 2020; Jakesch et al. 2023; Williams-Ceci et al. 2026), and suggested politeness changes writers' wording and their social perceptions (Zhang et al. 2026).

We therefore evaluate interventions with a ledger instead of a score. Our running case is keyword-to-sentence expansion with four LLMs on minimal-pair item families, built in concurrent work that specifies and tests meaning-preservation requirements and the scorer itself (Anonymous 2027). Here we ask a different question: *what does each intervention buy, what does it break, what does it cost, and where does it apply?*

Contributions. (1) An offline evaluation design for text-generation interventions: channel-controlled attribution, which decides what an intervention may be credited with; paired outputs of stored pipeline configurations; cluster bootstrap; an explicit harm ledger; and kill criteria fixed before the test run (§5). (2) Results led by what a gain-only report would hide: the guard's residual failures and partial passes; a menu intervention's collateral loss; and an applicability boundary between ambiguous and supplied input (§6). (3) A replication of the user-side cue on fresh items in which one of our own pre-specified criteria fails. (4) A *proposed* human-outcome protocol that these offline results cannot replace (§7). We ran no study with AAC users and claim no effect on communication, agency, effort or trust; every result is a change in stored text-system behavior on constructed items.

2 Related Work

LLMs in AAC. LLM expansion and suggestion systems for AAC include KWickChat (Shen et al. 2022) and SpeakFaster (Cai et al. 2024), and AAC users have described how

such systems help and hinder them (Valencia et al. 2023). Several add user-side control over intent: context and intent choices (Valencia et al. 2024; Mao et al. 2025), editable stance tags (Xiang et al. 2025), and retained function words such as "not" (Lacosse, Tano, and McNamee 2026). Others measure whether affect or intent survives generation (Choudhury, Kumar, and Martin 2025; Gorenshtein et al. 2026). Frisch et al. argue that AI-powered AAC needs technical, user-mediated and functional evaluation together (Frisch et al. 2026); our ledger is a technical layer shaped so that a functional study can test it.

Repair, preservation and suggestion sets. Generate–critique–revise loops are established (Madaan et al. 2023; Gou et al. 2024), and conditioning a rewrite on the source dialog act, then scoring act retention separately, already exists for style transfer (Qu, Ungar, and Sedoc 2023). We claim neither the loop nor act-aware rewriting. Our critic checks only what the user *supplied*, and we evaluate what the loop and the menu displace, not only what they fix. Models have default politeness strategies (Zhao and Hawkins 2025), socially oriented tuning can change other properties (Ibrahim, Hafner, and Rocher 2026), and writing with models can narrow what people write (Padmakumar and He 2024); this is why we score whole menus across all labeled dimensions.

Evaluating AI-mediated communication. Hancock et al. call for studying effects of AI-mediated communication on self-presentation and relationships (Hancock, Naaman, and Levy 2020). Opinionated co-writers shift users' views (Jakesch et al. 2023), biased autocomplete shifts attitudes (Williams-Ceci et al. 2026), AI replies change language and social perception (Hohenstein et al. 2023), and suggestions pull writing toward dominant styles (Agarwal, Naaman, and Vashistha 2025). Those are human-outcome studies of exposure; ours is a pre-deployment study of interventions meant to *prevent* unwanted shifts. LLM judges are common (Zheng et al. 2023) but can agree with each other more than with people on pragmatic labels (Wang, Sun, and Guo 2026; Panickssery, Bowman, and Feng 2024), which bounds what our machine labels can show.

3 Setting

Items and channels. Items are minimal-pair *families*: two source sentences that differ on one critical dimension ("I don't like the new medication" / "I like the new medication") and collapse to the same *bare* keywords (`like medication`). Each source is rendered through five input *channels*: *bare*; *noncrit*, which adds one non-critical word as a length control; *chip1*, which adds the critical cue (`[NOT] like medication`); *chipall*, with cues for all commitments; and *full*, the typed source (`The user typed: ...`). We use three populations from the concurrent study (Anonymous 2027). The *discovery benchmark* has 58 held-out families on six dimensions (polarity, agent, certainty, speech act, politeness, time/quantity), after quarantining two refusal/acceptance families whose scoring rule has a known blind spot; all guard and I2 results use it. *W-C* has 240 fresh families (polarity, speech act, politeness), 238 analyzed after a pre-generation exclusion, with three registered contrasts; its confirmation is qualified because 48 cached requests (12 distinct primary ones) predate the final item freeze. *W-D* has 80 physical scenarios that add a second cue to a satisfied one. Families are constructed, not messages written by AAC users.

Attribution decides what an intervention may claim. Because the channel fixes which commitments the model received, every miss is attributed by construction (Anonymous 2027). *Loss*: the channel never carried the value. *Violation*: the channel carried it and the output changed it. *Assumption*: the channel was ambiguous and the output committed anyway. A checker can only enforce a commitment the user supplied, so the guard is credited with Violations alone. When the keywords leave a dimension open, no checker can know the intended reading; the remedies are to show alternatives (I2) or to let the user supply the cue (chip).

Systems and scoring. Systems are Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct and Gemma-3-4B-it (4-bit, llama.cpp) and the API model gpt-6.1-terra. P0 asks for one sentence (greedy); P2 asks for three suggestions that "cover the different things the user might mean"; S3 draws three samples at temperature 0.7. An LLM extractor (gpt-6.1-sol, two prompt variants) labels each output's dimensions without seeing source, model or arm; a cross-family arbiter resolves critical-label disagreements, and deterministic code computes verdicts. Treated and untreated outputs therefore pass through the same scoring. In W-C, 259 disagreement rows (177 units) were resolved by a single arbiter after outcomes were known, blind to reference, model and arm; the earlier analysis that masked them as unknown gives the same three conclusions. Against two human annotators on 163 sampled untreated outputs, extractor labels agree at held-out $\kappa = 0.91$–0.97 on all 7 label fields (lowest 95% bound 0.81; pre-specified gate 0.6), and Violation verdicts recomputed from human labels match on 96/99 units ($\kappa = 0.92$). That gate covers extraction on untreated outputs; it does not validate repaired or rewritten text.

4 Interventions

I1: commitment guard (system side, runtime). After the first-pass P0 output, a deterministic lexical monitor (regexes and lexicons for negation, person, modality, question form, numbers, tense and "please") compares the output with the commitments in the visible input: the chips, or for full input the monitor's own labels of the typed sentence. It also flags a politeness marker that was not supplied. On a flag, the same conversation receives one feedback turn ("Your sentence must keep: negated; no added politeness. Write the sentence again with the same meaning otherwise.") and is regenerated, up to two retries. The guard returns the first unflagged attempt, else the attempt with fewest flags, ties going to the original. The monitor never sees the evaluation labels. *G1s* is a zero-call comparator that only deletes an unsupplied "please".

I2: ambiguity-complete suggestion set (system side, bare input). Starting from the P2 set, I2 keeps the top suggestion. For polarity and then certainty, if the monitor finds

no slot carrying the alternative reading, I2 rewrites the third (then second) slot into that reading with one LLM call ("make it UNCERTAIN: a statement that is not sure (use maybe, might or I think)"), verified by the monitor with one feedback retry. I2 was chosen over a variant that rewrites the top suggestion (I2t) on a 12-family development pilot before the test run.

Chip (user side). The user adds one commitment cue, such as `[NOT]`, `[MAYBE]`, `[?]` or `[PLAIN]`, compared with a length-matched non-critical word at a nominal cost of one selection.

5 Evaluation Design

Table 1 gives the design in the form the workshop asks for: intervention, comparison, population, outcomes and time horizon.

Paired comparisons of stored configurations. For the system-side interventions, the untreated outcome is the *stored* first-pass output for the same request, and the treated outcome is what the intervention returns from it. The contrast compares paired outputs of two stored pipeline configurations; it removes between-request variation but does not identify an effect on people, and it does not separate regeneration from serving differences. Pairing matters because greedy decoding is not reproducible: issuing the same request twice reproduced the text in only 76.4–90.8% of local and 44.4–71.4% of API requests. The chip comparison pairs chip1 with noncrit for the same target; these are different requests, so run-to-run churn is part of that contrast.

Estimands and uncertainty. Estimands are row-weighted means over targets or sets; the four models are fixed, not sampled. Discovery intervals are 95% percentile intervals from 2,000 whole-family resamples (seed 20261031); W-C and W-D use 10,000 resamples of recorded seed/construction clusters or physical scenarios (seed 20261031); the exploratory applicability contrasts use 10,000 cluster resamples (seed 20261101). Intervals are pointwise and do not correct for exploratory selection or label error; a zero-event interval is not a guarantee. Thousands of rows are not thousands of independent tests: the guard's 63/80 repairs come from 26 families.

Harm ledger and cost. Beside each benefit we report (i) *changed-correct*, already-correct outputs whose text the intervention changed, a potential unwanted edit even when (ii) *pass→fail*, the share that became scored failures, is zero; (iii) *residual* failures; (iv) *collateral coverage loss* on dimensions the intervention did not target; and (v) *act drift*, rewrites that change sentence type. The ledger also records what a label cannot see, such as scored passes that are fragments. We count extra generator calls and wall-clock latency; local latency was measured with partial CPU offload in this serving setup and is not a deployment prediction.

Kill criteria. Each intervention had a pass/fail rule (Table 1) checked on a 12-family development pilot before the held-out run; a failing intervention would be reported as killed, not tuned on test. The rules are coarse by design: they decide whether an intervention merits a human study, not whether it is good.

6 Results

6.1 I1: the guard repairs most violations, leaves some, and a pass is partial

On cue channels, strict pass (critical value kept, no supplied dimension contradicted, no politeness added) rose from 1,312 to 1,375 of 1,392 targets (94.25% to 98.78%), an absolute gain of 4.53 points (Fig. 1). Among the 80 initial failures, the guard repaired 63 (78.8% [68.4, 88.9]), far above its 30% kill line; the zero-call strip-only comparator repaired 34/80, all politeness additions.

Residual failures. 17 of 80 failures remained (17/1,392 cue targets, 1.2%). 12 of them involve speech act, and 14/17 were still flagged by the monitor after both retries: detected, not repaired. In the gallery (Fig. 2), the guard regenerates "Can you charge my wheelchair?" for `[?] [YOU] charge wheelchair` and returns the same request, although the source asks whether the listener is charging it. A retry loop is not a guarantee even when the monitor notices the error.

A pass is only partial. The scorer checks a handful of labeled dimensions, so some scored repairs are not usable sentences. For `[ASK] [YOU] [PLAIN] wear headphones`, the guard turns "You wear headphones." into "Ask you to wear headphones.", which passes; 4/63 scored repairs begin with this "Ask you" chip echo. That count is a post hoc lexical diagnostic, not a fluency rate. No independent human audit of repaired text has been completed, so we report *scored predicate repairs*, not usable repairs. The monitor and the evaluator are separate programs that measure the same constructs: on first passes the monitor's flag agreed with judged failure at $\kappa = 0.87$ (precision 0.80, recall 0.96), so a repair is partly a repair as seen by an extractor of the same kind.

Harms and cost. The guard changed the text of 49/2,154 already-correct outputs (2.3% [1.1, 3.6]), inside its 10% kill line, 14 of them on cue channels; none became a scored failure. Zero scored failures does not mean no unwanted edit: "Would you mind repeating the last question" becomes "Please repeat the last question", still polite by the label and no longer what was typed; harm to people was not measured. The guard flagged 157/1,856 inputs and added 0.104 generator calls and 0.17 s per input. For typed sentences in an explicitly selected full-sentence mode, returning the input verbatim passes the same checks for 116/116 authored sources at zero calls; this is a baseline, not an expansion method.

6.2 I2: completing a menu displaces other readings

Under bare input, P2 menus offered both polarity or both certainty readings in 15/80 sets; I2 raised this to 78/80, a paired gain of 78.8 points [70.0, 87.5], far above its kill line (three samples, S3: 1.2%). The slots I2 rewrites are the slots that

	I1 commitment guard	I2 ambiguity-complete set	Chip (user side)
Defect targeted	Violation of a supplied commitment	Assumption under ambiguous input	Loss of a cue at input
Comparison	Stored P0 first pass, same request; G1s strip-only (0 calls)	Stored P2 set, same request; S3 samples	Length-matched non-critical word (noncrit), same target
Population	Discovery: 1,392 cue targets, 58 families	Discovery: bare-input sets of 3, one per family and model	Discovery: 464 targets; W-C speech act: 624 targets, 78 families
Primary benefit	Share of scored violations repaired	Sets offering both readings, polarity and certainty families	Critical-value match, chip1 vs. noncrit
Harm ledger	Correct outputs changed; correct→failure; violations left; scored passes that are fragments	Coverage lost on non-targeted dimensions; sentence-type (act) drift of rewrites	Matches lost when the chip is added; misses not repaired; gap to full input
Applicability	Supplied cues only (chips, typed sentence)	Ambiguous input only; on supplied input, diversifying departs from it (§6.3)	Dimension-specific; weakest for speech act
Cost	Generator calls, latency, monitor time	Generator calls, latency	Nominal selections (reading time not counted)
Kill criterion	Repair $\geq$30% of violations *and* change $\leq$10% of correct outputs: passed	Gain $\geq$30 points over P2: passed	Gain $\geq$10 points over noncrit *and* $\geq$80% of full-input match: discovery passed; W-C speech act failed (gain passes, 80% clause fails)
Human outcome (proposed)	Message sent as meant; corrections; time to send	Suggestion accepted; meaning endorsed	Chip use; message sent as meant
Time horizon (proposed)	Per message; weeks	Per set; weeks	Per message; weeks

Table 1: Evaluation card. Rows down to "Kill criterion" are evaluated offline, with the verdicts computed from stored estimates (Fig. 1 shows the same verdicts). The last two rows belong to the proposed human study (§7); nothing about them is measured. Kill criteria were fixed in the analysis code and checked on development data before the held-out run; they were not externally timestamped. Guard and I2 have not been run on W-C.

carried other readings. On the separate politeness families, both-reading coverage fell from 26/40 to 12/40 sets, 35.0 points [15.0, 57.5]; speech-act coverage fell by 25.8 points [9.7, 40.6]. Target gains and politeness losses are measured on different family strata, not as a within-item trade. In the gallery's `knock come` menu, "Please knock before you come in." becomes "Don't knock before you come in.", removing the polite option, and "Did someone knock before coming in?" becomes a "Maybe" statement. Of 428 rewrites, 138/428 (32.2%) change speech act, more often for certainty than polarity edits (96/220 vs. 42/208); the instruction asks for "a statement", which a request cannot satisfy without changing type, though slots receiving each edit differ, so this is descriptive. Across all six dimensions coverage still rose 17.6 points [4.0, 31.6], at 2.05 extra calls and 3.25 s per set. Coverage is not faithfulness: gained readings can invent details the keywords did not contain. An applicability-aware rewrite that may decline (T3) was specified with its own kill rule but has not been run; it is deferred.

6.3 Applicability: diversify ambiguous input, preserve supplied input

I2 assumes that the input leaves a reading open. W-C also generated P2 menus for inputs that do *not* leave the critical value open, which lets us test what diversification does there (Table 2). For a typed sentence, a P2 menu contained at least one output whose focal label departs from the supplied value in 62.76% of targets, against 2.70% for three samples of the same size: 60.05 points [57.31, 62.70]. Excluding the families with early cached requests leaves 59.92 points. A narrower judge-free endpoint confirms the direction: 21/639 full-input polarity menus contain the *other* authored source sentence verbatim after normalization, versus 2/639 sample sets (2.97 points [1.55, 4.54]). In the gallery's last row, "I don't have a doctor's appointment on Tuesday morning." is typed, and the menu's second suggestion is "I have a doctor's appointment on Tuesday morning." The categorical rate is a partial-label endpoint, not an opposite-meaning rate, and the comparison does not control prompt or joint versus separate generation. Its engineering consequence is still direct: *diversifying readings that keywords leave open and preserving*

Channel	P0 (one)	P2 (three)	S3 (three)	P2−S3, pp
full	2.05	62.76	2.70	60.05 [57.31, 62.70]
chip1	12.09	38.67	12.91	25.76 [23.69, 27.88]
chipall	13.21	37.43	14.46	22.98 [21.38, 24.64]

Table 2: W-C supplied-input channels: % of targets whose output or menu contains at least one focal-label mismatch with the supplied value (full: 1,161/1,850 P2 vs. 50/1,850 S3 complete matched targets of 1,904; 210 clusters). Exploratory comparison outside W-C's registered contrasts; categorical mismatch is not necessarily opposite meaning. P0 has one opportunity, so it is a diagnostic only.

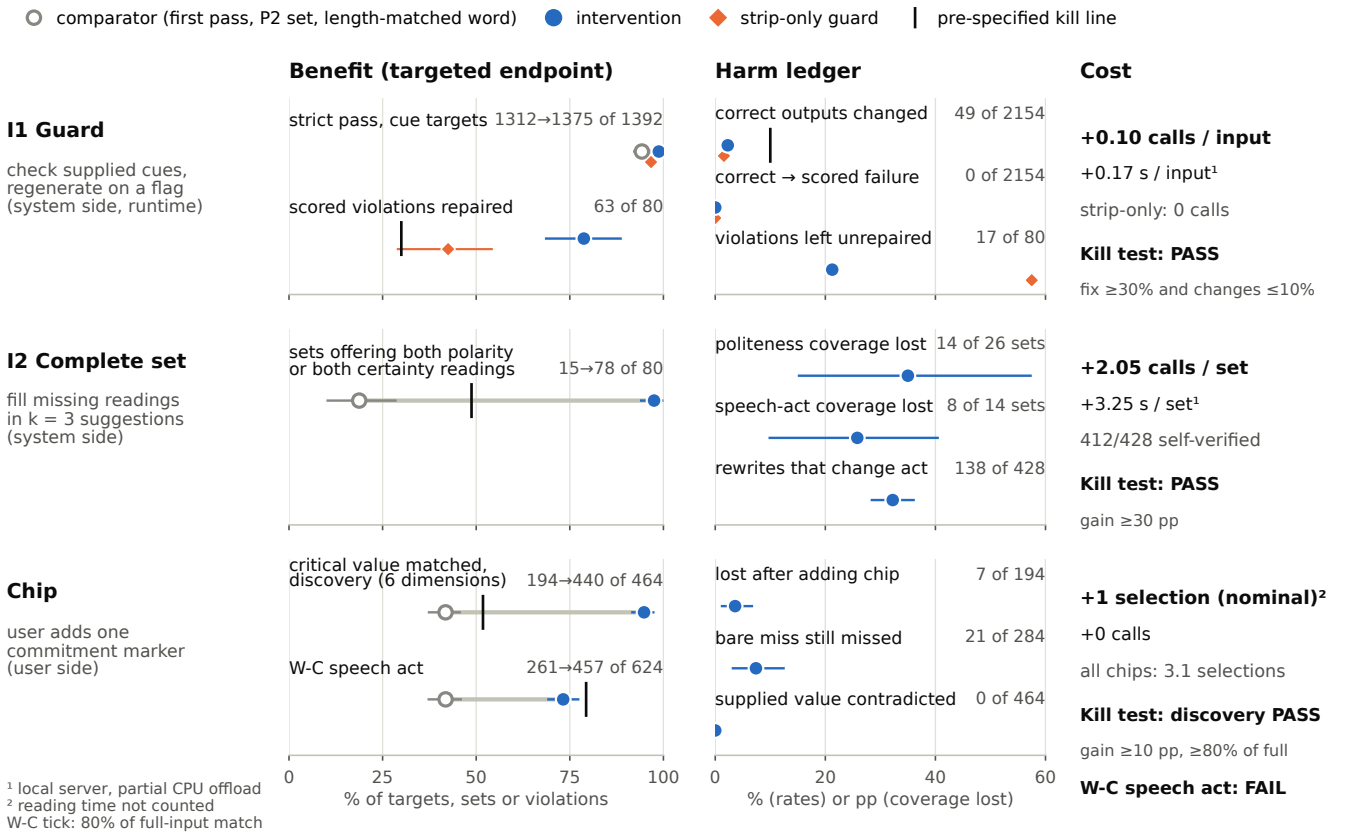


Figure 1: Benefit, harm and cost for each intervention (machine-extracted labels; four models). Guard and I2 rows use the 58-family discovery benchmark. The chip row shows the discovery result (six dimensions) and the W-C speech-act replication on fresh families, whose tick marks 80% of full-input match; W-C passes the gain clause but fails that clause. Open circle: comparator; filled: intervention; diamond: zero-call strip-only guard. Coverage losses are points against the P2 set. Bars are 95% cluster-bootstrap intervals; counts give observed numerators and denominators. Kill verdicts are computed from the estimates and match Table 1.

values the input fixes are different requirements, so a system should condition diversification, including I2, on which dimensions are actually open. I2 itself was not run on supplied input.

6.4 Chip: replicated on fresh items, still far from full input

On the discovery benchmark, one chip raised critical-value match from 41.8% [37.1, 45.9] with the length-matched word to 94.8% [91.4, 97.6] (97.3% of full input), passing both kill clauses; speech act was the weakest dimension (81.2%). W-C tested that weakest case on 78 fresh speech-act families as a registered contrast. The speech-act chip raised preservation from 41.83% to 73.24%, 31.41 points [25.00, 37.94] over 624 paired targets in 55 clusters (Holm $p = .0003$), so the gain clause replicates. The criterion does *not* transfer: full input preserved speech act in 99.20%, so one chip reaches 73.8% of the full-input level, below the $\geq$80% clause, with a paired chip1$-$full gap of -25.96 points [-30.36, -21.53]. For polarity, by contrast, chip1 matched 99.22% against 99.84%. Question cues fail most visibly: 58/208 question targets became non-questions under chip1, as when `[?] parcel signature` for "Does the parcel from the bank need a signature?" became "Can you sign for this parcel?" (Llama). The nominal cost of one selection is a lower bound that shifts the burden of each non-default meaning onto the user (Frisch et al. 2026). W-C's other registered contrasts also passed: typed plain requests gained politeness in 26/320 outputs and lost it in 6/320 (difference 6.25 points [2.50, 10.00]), and bare keywords were expanded affirmatively in 99.37% of directional outputs; both are the concurrent study's subject (Anonymous 2027) and motivate, but do not test, the interventions here.

6.5 Composition: a second cue can undo a satisfied one (W-D)

Adding a second cue is itself an intervention on the input. In W-D, Llama under a transparent `[STATEMENT]` encoding lost an already-preserved negation in 55/640 aligned transitions (8.59% [6.25, 11.09]) across 30/80 scenarios, against 0/160 same-input repeat disagreements; all occur when `[NOT]` meets `[YOU]` (55/160; other cells 0/480),

Arm	Visible input	Before (comparator)	After (intervention)	Model, item, declared rule, rate of that rule's population
I1 repaired	`[PLAIN] let finish sentence`	"Please let me finish my sentence."	"Let me finish the sentence."	Gemma, pol2-14/A, chip1. Rule: first by hash order. 63/80 scored failures repaired.
I1 repaired	`[ASK] [YOU] [PLAIN] wear headphones`	"You wear headphones."	"Ask you to wear headphones."	Llama, pol2-07/B, chipall. Rule: second by hash order. Scored repair that is a fragment: 4/63 scored repairs start "Ask you" (post hoc lexical count, not a fluency rate).
I1 not repaired	`[?] [YOU] charge wheelchair`	"Can you charge my wheelchair?"	"Can you charge my wheelchair?"	Llama, act-11/A, chipall. Rule: first by hash order. 17/80 failures remain; 12/17 involve speech act; 14/17 still flagged by the monitor.
I1 changed correct	`The user typed: Would you mind repeating the last question?`	"Would you mind repeating the last question."	"Please repeat the last question."	Llama, pol2-13/A, full. Rule: first by hash order. Already-correct cue targets changed: 14/1312; none became a scored failure, yet the softener form changed.
I2 coverage gain	`smell gas`	1. "I smell gas in the kitchen." [plain] 2. "There is gas smell coming from the basement." [plain] 3. "Do you smell gas outside?" [?]	1. "I smell gas in the kitchen." [plain] 2. "There might be a gas smell coming from the basement."† [hedged] 3. "Do you not smell gas outside?"† [?]	Qwen, cer-11. Rule: first polarity/certainty set gaining both readings (hash order). Sets with both readings 15/80→78/80; 63/65 uncovered sets gained. Slot 3 also changed and locations were invented.
I2 collateral loss	`knock come`	1. "Someone knocked and came in." [plain] 2. "Did someone knock before coming in?" [plain] 3. "Please knock before you come in." [polite]	1. "Someone knocked and came in." [plain] 2. "Maybe someone knocked before coming in."† [plain] 3. "Don't knock before you come in."† [plain]	Terra, pol2-08. Rule: first politeness set losing both readings (hash order). Politeness coverage 26/40→12/40; 14/26 covered sets lost. Different families from the row above.
I2 act drift	`bring dog`	"Can you bring the dog?"	"I think you might bring the dog."	Llama, act-07, slot 2 (certainty rewrite). Rule: first certainty edit that changes act (family, model, slot). Certainty edits changing act: 96/220.
Full-input set	`The user typed: I don't have a doctor's appointment on Tuesday morning.`	P2: 1. "I don't have a doctor's appointment on Tuesday morning." 2. "I have a doctor's appointment on Tuesday morning."‡ 3. "Do you have a doctor's appointment on Tuesday morning?"	S3: 1. "I don't have a doctor's appointment on Tuesday morning." 2. "I do not have a doctor's appointment on Tuesday morning." 3. "I don't have a doctor's appointment on Tuesday morning."	Llama, r2pol-024/B, full input. Rule: first full-input P2 set containing the other authored source (‡) (family, model, variant, slot order). Sets reproducing the partner source: P2 21/639 vs. S3 2/639.

Figure 2: Before/after gallery of stored outputs, verbatim. Each row is chosen by the outcome-conditioned rule it names (hash order is SHA-1 of request, family, variant and channel, as in the intervention report); its rate describes that rule's population, so the gallery is illustrative, not a random sample. Set rows show all three suggestions in both arms, in slot order, with the scored reading ([?] = unclear on that dimension); † marks slots I2 rewrote. Guard and I2 rows come from the discovery benchmark; the last row is W-C (exploratory channel comparison), whose two columns are the P2 and S3 arms for the same typed sentence rather than before and after. Labels are machine-extracted; the human extraction gate covered untreated outputs, not these repairs.

as in `[NOT] [STATEMENT] [YOU] lock door` → "You lock the door." Other models showed no polarity damage in that encoding, and a Qwen agent effect appears only under another encoding with a known codebook limitation. This supports per-model, per-encoding regression tests for cue combinations, not model-general interference; constraint order is an unresolved contributor (Zeng et al. 2025).

6.6 Reading the ledger across interventions

Each intervention answers a different defect, and the ledger keeps them apart. The guard enforces supplied commitments at about a tenth of a call per input; its harms are edits to correct outputs, residual act failures and scored passes that are fragments. The completed menu exposes readings the input left open at about twenty times the guard's call cost; its harms land on dimensions it did not target, and on supplied input diversification departs from what the user supplied. The chip moves work to the user and is cheapest per system call, but for speech act it remains well short of the typed sentence. Effects differ by model: Gemma's first pass is worst and the guard repairs 29/33 of its violations against 7/13 for Qwen, and I2's act drift ranges from 13/110 (Qwen) to 44/102 (Gemma) rewrites, so a deployment decision needs the ledger for the model it ships. A gain-only report would have credited the guard with all channels' violations, I2 with a 78.8-point gain, and the chip with a passed criterion that fails on fresh speech-act items.

7 A Proposed Human-Outcome Protocol

The results above are effects on stored text. Whether they become effects on people is a causal and longitudinal question, and the answer may differ: a guard that repairs most violations might teach users to stop marking negation, and menus that offer "maybe" might shift how certain a speaker sounds over weeks. We sketch, as a proposal only, a study that would test this; it has not been run, approved by an ethics board, or designed with AAC users, and doing so with them is a precondition, not a detail.

Design and estimands. A micro-randomized trial (Klasnja et al. 2015) with AAC users who already compose with keyword expansion, over several weeks of their own use, would treat each expansion request as a decision point and randomize only where the arms differ, which the offline ledger identifies (e.g., the guard's output vs. the original at the 8.5% of inputs the monitor flags; I2 vs. P2 menus for bare keywords). Chips are the user's choice, so only their availability could be randomized, an encouragement design whose estimand differs from the effect of using a chip. The primary estimand would be the causal excursion effect (Boruvka et al. 2018) on proximal outcomes (sent without edit, corrections, time to send, an occasional one-tap "said what I meant"), with moderators such as channel, flagged dimension and time in study. Randomization alone does not settle consent for partner-side logging, carry-over to later messages, dependence on treatment history, or habituation; each needs its own design decision and lagged or longitudinal outcomes (e.g., chip use and plain/polite share over weeks).

What the offline ledger contributes. It narrows the study to decision points where arms differ, names the harms worth measuring (unwanted edits, displaced readings, residual act errors), and supplies falsifiable surrogate hypotheses: if offline changed-correct or coverage loss predicts "said what I meant", the cheap ledger can screen later interventions; if not, evaluation of this kind should be demoted.

8 Limitations

No human outcomes. Nothing here measures communication success, agency, effort or trust. *Machine labels.* Endpoints use an LLM extractor and deterministic verdicts. Its human validation used two annotators on 187 items (a double-coded core with $\alpha = 0.95$–1.00); labels were converted from another format, ties were excluded rather than adjudicated, and agreement is near ceiling on a codebook for explicit markers. It covers untreated outputs only: repaired and rewritten text and new W-C/W-D boundary cases are not human-validated, and shared monitor/extractor errors can inflate guard repairs. *Scope of each result.* Guard and I2 are discovery-benchmark results; neither has been replicated on W-C. W-C's confirmation is qualified by early cached requests and by post-outcome arbitration of disagreement labels. The applicability contrasts are exploratory channels outside W-C's registered family, and W-D lacks a complete prospective item/codebook freeze. *Constructed items, fixed models.* Families are authored; four fixed models describe these systems, not AAC users' messages or other models. A corpus-transfer study is reported in the concurrent work (Anonymous 2027). *Narrow endpoints.* Labels cover six coarse dimensions; a scored pass can be a fragment, and invented details are not penalized in coverage. *Selection.* I2 was chosen over I2t on development data; gallery rows follow declared, outcome-conditioned rules and illustrate rather than sample. Act-drift and residual decompositions were specified after earlier results; their intervals are not corrected for that search. *Serving.* Local intervention calls ran with partial CPU offload; latency is comparable only within this study.

9 Conclusion

Each intervention passed its pre-specified test on the discovery benchmark, and each carries a measured cost that a gain-only report omits: residual failures and fragmentary scored passes for the guard, displaced readings for the completed menu, and, on fresh items, a speech-act cue that improves preservation yet stays far below the typed sentence and fails one of our own criteria. The applicability result adds a boundary: diversifying readings is a remedy for ambiguous input and a source of departures from supplied values once the input fixes them. Reporting benefit, harm, cost and applicability together turns a ranking question into design questions that a human study with AAC users can test. We release the analysis code, stored outputs and scripts that rebuild every number and figure from them.[PENDING: anonymized artifact URL]

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